

GISELLE WU

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EDUCATION

Dartmouth College, Hanover, NH

June 2027

Bachelor of Arts in Computer Science Modified With Engineering, Bachelor of Engineering

GPA 3.75/4.0

Relevant Coursework: Algorithms and Data Structure, Theory of Computing, Computer Graphics, Full-Stack Development, Digital Electronics, Software Design and Implementation, Problem Solving via Object-Oriented Programming, Differential Equations

AIT Budapest, Budapest, Hungary

September - December 2025

Bachelor of Computer Engineering

Relevant Coursework: Algorithms and Data Structures, Mobile Software Development, Theory of Computing, Computer Graphics

TECHNICAL SKILLS

Languages: Python, Java, C/C++, MATLAB, R, JavaScript, Kotlin, SQL

Frameworks/Tools: React, Node.js, Git, Linux, REST APIs, Flutter

Hardware & Design: FPGA (Basys3), VHDL/Verilog, VGA Controller Design, Digital Logic, CAD (SolidWorks, CSWA)

Data & Analytics: Data Structures & Algorithms, Machine Learning, Statistical Analysis, Data Modeling, Excel, PyTorch

UI/UX & Web: Responsive Design, Figma, DOM Manipulation

RESEARCH & WORK EXPERIENCE

Jacobson Lab, Hanover, NH

May 2025-Present

Undergraduate Research Assistantships at Dartmouth (URAD) Intern

- Developed and integrated the first text-to-speech (TTS) modules into a cross-platform AI chatbot, enhancing conversational flow and user engagement.
- Built a cross-platform chatbot in Flutter, implementing both frontend UI and backend logic for real-time, smooth performance.

Makerspace, Hanover, NH

May 2025-Present

Trainee

- Co-led hands-on training sessions on 3D printing, laser cutting, and sewing for 20+ peers per term
- Supported a collaborative, creative environment by mentoring new makers and troubleshooting fabrication challenges
- Gained practical expertise in digital fabrication and rapid prototyping, fostering innovation across student projects

Lynch Rocket Lab, Hanover, NH

January 2024-January 2025

Data Analytics Intern

- Processed Swarm-Over-Poker satellite and radar data using replication method on MATLAB
- Engineered a new computational grid for Discovery by implementing parallel processing techniques, optimizing data preparation for GEMINI simulations on NASA's Pleiades Supercomputer, resulting in a 60% reduction in computation time.

SELECTED PROJECTS

Pet Rock VGA Game

August 2025

- Designed and implemented **5+** RTL state machines in VHDL for a Tamagotchi-style FPGA game
- Integrated and verified real-time VGA output (640×480, 60 Hz) through full simulation and hardware testing in Vivado and oscilloscope
- Debugged **2000+** lines of VHDL to achieve stable gameplay; awarded **Top Score** for Innovation out of 20+ projects

Harmonize

May 2025 - June 2025

- **Built a full-stack interactive map** using Leaflet to display nearby friends and events via geolocation, reverse geocoding, and the Ticketmaster API; collaborated in a team of 5
- Implemented "Blend Mode" to calculate music compatibility (shared artists/genres); used React, Zustand, API integration, and debounced async calls for efficient, real-time updates.
- Beta-tested with 20+ users, improving event discovery and music compatibility matching

Walking Dino

May 2025 - June 2025

- Designed, built, and tested a mechanical walking dinosaur using a four-bar linkage system modeled in SolidWorks, achieving 100% walking stability over **20+** test trials
- Fabricated over 15 custom parts using 3D printing and laser cutting, resolved all mechanical interferences, and achieved smooth, repeatable locomotion

LEADERSHIP EXPERIENCE

Alpha Xi Delta Sorority, Hanover, NH

January 2025 - Present

Communications Vice President

- Represented 300+ members in media outreach; produced newsletters and branded gear, strengthening alumni engagement.

Sheba Dance Troupe, Hanover, NH

September 2023 - Present

Logistical Director

- Directed logistics for 350+ audition participants across 5 teams, handling scheduling and inter-team communication.